南京航空航天大學
NANJING UNIVERSITY OF AERONAUTICS AND ASTRONAUTICS

FT-GAN: Fine-Grained Tune Modeling for Chinese Opera Synthesis

Meizhen Zheng[1,2*], Peng Bai[1,2*], Xiaodong Shi[1,2†], Xun Zhou[1], Yiting Yan[1]

[1]Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen, China
[2]Key Laboratory of Digital Protection and Intelligent Processing of Intangible Cultural Heritage of Fujian and Taiwan (Xiamen University), Ministry of Culture and Tourism, China
{midon, baipeng}@stu.xmu.edu.cn, mandel@xmu.edu.cn, {xzhou, eatingyan}@stu.xmu.edu.cn

AAAI 2024

Background

- 大背景：戏曲相关的数据集少，相关研究少
- 单字音多音高（跑腔）　　例：a a a / A3 B3 D4
- 音素时长变化多样
- 元音变形　　例：ai变a ai i

Dataset Construction

文件 编辑 查看

sp ua k ah sp a ts i i sp p ut k ang b u u kh iok s i k
0.53 0.395 0.032 0.432 0.232 0.35 0.091 0.298 0.364
0.095 0.206 0.124 0.309 0.041 0.534 0.213 0.022 0.7
sp G4 G4 G4 sp E4 G4 G4 C#4 sp F4 F4 C4 C4 A#4 A
0 1 0 0 0 1 0 0 1 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0
0 1 0 0 0 1 0 0 1 0 0 1 0 1 0 0 1 0 1 0 1 0 1 0 0 1 0

GOAT数据集示例

Methods

Model composition

- Acoustic Model： conditioner， predictor， discriminators

- vocoder： HifiGAN

Methods

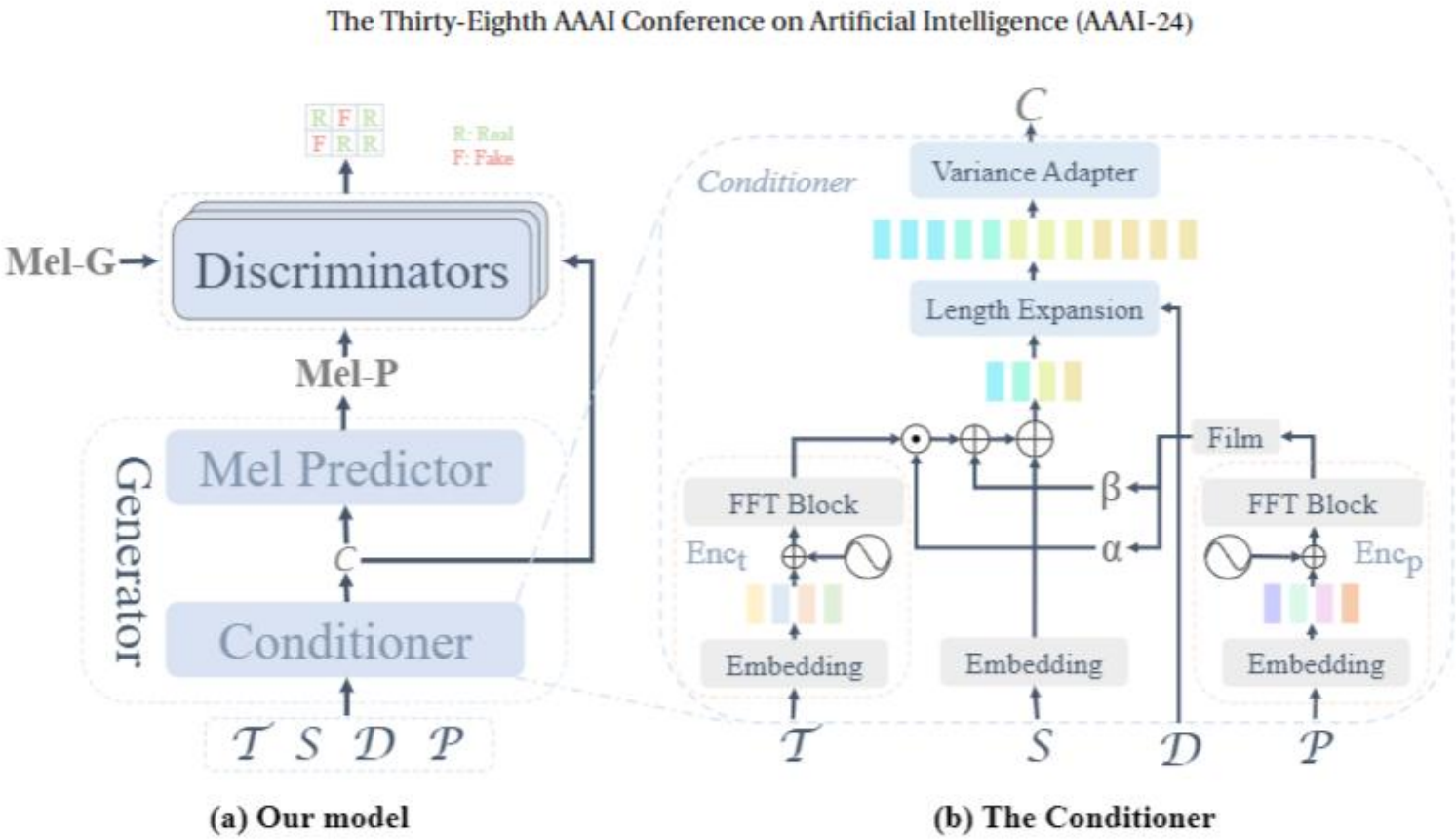


Figure 1: (a) The overall architecture of the model, where Mel-P indicates the predicted mel-spectrogram and Mel-G indicates the ground-truth mel-spectrogram. $\mathcal{T}$, $\mathcal{S}$, $\mathcal{D}$, and $\mathcal{P}$ are the phoneme sequence, the speaker label, the duration sequence, and the pitch sequence. $\mathcal{C}$ is the unified conditional representation of these four conditional inputs. (b) The architecture of the conditioner, where Enc_t indicates the phoneme encoder, and Enc_p indicates the pitch encoder. The film is an information-integrating module.

Methods

$$L_{total} = L_1 + \omega L_G + L_D + \alpha_p L_{per} + \alpha_f L_{fm}$$

where α_p, α_f represent different loss weights.

$$L_G = \mathbb{E}_{\hat{y}}[\sum_{i\in\{l,m,h\}} \rho_i(1 - D_i(\hat{y},\mathcal{C})) + \sum_{t\in\{s_1,s_2,s_3\}} \frac{1}{3}(1 - D_t(\hat{y},\mathcal{C}))] \quad (1)$$

$$L_D = \mathbb{E}_{y,\hat{y}}[\sum_{i\in\{l,m,h\}} \rho_i(1 - D_i(y,\mathcal{C}) + D_i(\hat{y},\mathcal{C})) + \sum_{t\in\{s_1,s_2,s_3\}} \frac{1}{3}(1 - D_t(y,\mathcal{C}) + D_t(\hat{y},\mathcal{C}))] \quad (2)$$

$$\hat{y} = G(\mathcal{P},\mathcal{T},\mathcal{S},\mathcal{D}) \quad (3)$$

where D_l, D_m, and D_h are three SF-GAN discriminators that handle the low, medium, and high-frequency regions of the spectrogram, respectively. ρ_l, ρ_m, and ρ_h are the corresponding weights. Discriminators D_{s_1}, D_{s_2}, and D_{s_3} are three ML-GAN discriminators that handle three different lengths of segments of the spectrogram. $\mathcal{C}$ is the output of the conditioner. We add another feature matching loss L_{fm} to assist the training (Chen et al. 2020):

$$L_{per} = \mathbb{E}_{y,\hat{y}}[\frac{1}{L}\sum_{l\in[1,L]} \|C^l(y) - C^l(\hat{y})\|_2^2] \quad (5)$$

where L represents the number of intermediate layers in the classification network, and C^l represents the normalization feature of the l-th intermediate layer in the network.

$$\omega = \frac{\|d(L_1 + L_{per})/dw\|_2^2}{\|d(L_G)/dw\|_2^2 + \epsilon} \quad (6)$$

where w represents the parameters of the last layer in the generator and ϵ is a positive value close to zero.

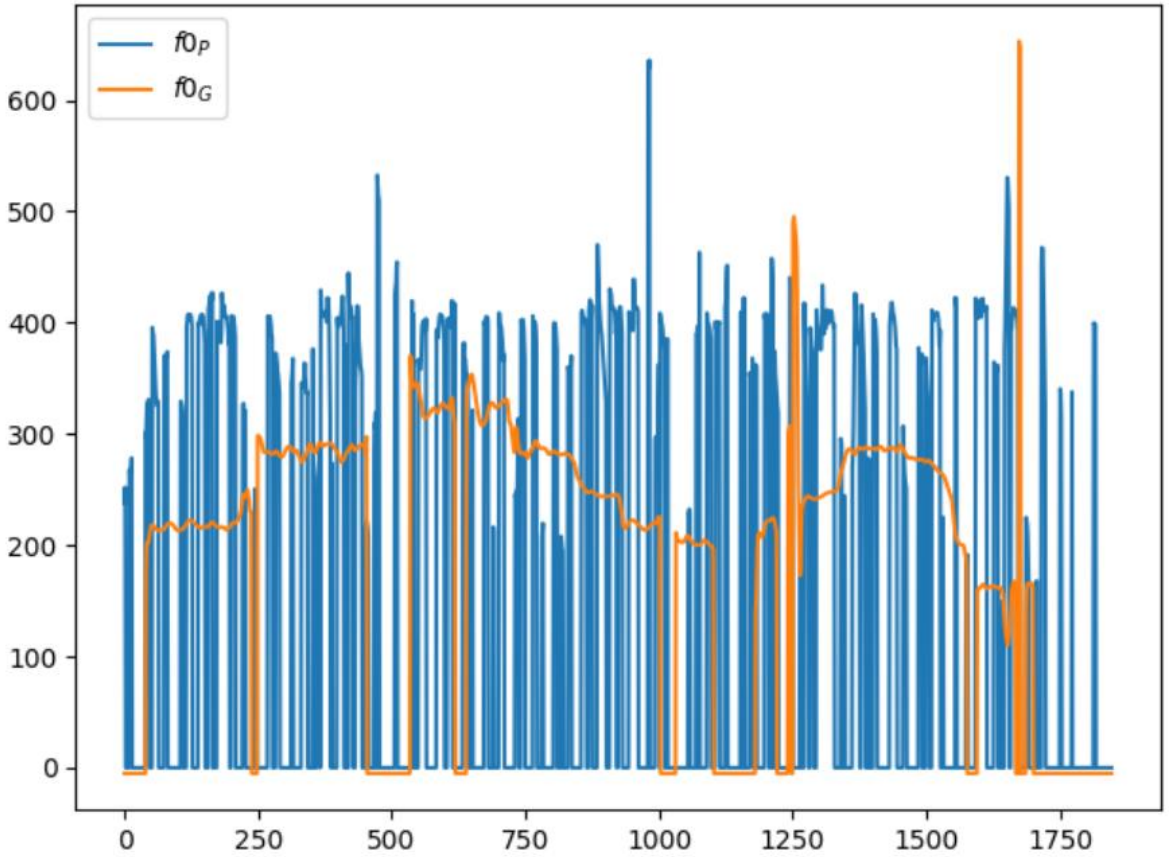
$$L_{fm} = \mathbb{E}_{y,\hat{y}}[\frac{1}{N}\sum_{i\in\{l,m,h\}}\sum_{l\in[1,N]} \rho_i \|D_i^l(y,\mathcal{C}) - D_i^l(\hat{y},\mathcal{C})\|_2^2 + \frac{1}{N}\sum_{t\in\{s_1,s_2,s_3\}}\sum_{l\in[1,N]} \frac{1}{3}\|D_t^l(y,\mathcal{C}) - D_t^l(\hat{y},\mathcal{C})\|_2^2] \quad (4)$$

where D_i^l (D_t^l) represents the normalization feature of the l-th of N intermediate layers in the i-th (t-th) discriminator.

Experiments

Model	Pro	Nat	Exp
GT	4.54 ± 0.03	4.30 ± 0.03	4.44 ± 0.03
FastSpeech2	4.26 ± 0.04	3.54 ± 0.04	3.76 ± 0.04
DiffSinger	3.96 ± 0.04	3.26 ± 0.04	3.52 ± 0.04
Visinger2	4.44 ± 0.03	3.90 ± 0.04	4.14 ± 0.04
FT-GAN	4.40 ± 0.03	$\mathbf{4.18 \pm 0.03}$	$\mathbf{4.22 \pm 0.04}$
-w/o SP	$\mathbf{4.46 \pm 0.03}$	4.08 ± 0.03	4.20 ± 0.03

Table 2: MOS test result with 95% confidence interval of the ground truth and different models on Gezi Opera synthesis. Pro indicates pronunciation, nat indicates naturalness, exp indicates expressiveness, and SP indicates speech pre-training.



Experiments

消融实验

Method	Pro	Nat	Exp	Con
w/o Vowel Morphing	-0.43	-0.36	-0.23	-0.18
w/o Vocal Run	-0.24	-0.39	-0.27	-0.32
w/o Pitch Encoder	-0.20	-0.34	-0.37	-0.25
w/o Discriminator	-0.23	-0.23	-0.11	+0.08
Conformer2FFT	-0.26	-0.32	-0.39	-0.26

Table 3: CMOS (Comparative Mean Opinion Score) test results of the ablation studies on annotation strategies and model structures. Conformer2FFT indicates replacing Conformer with FFT block in the mel predictor and con indicates consistency.

FTGAN在京剧上的表现

Model	Pro	Nat	Exp
GT	4.53 ± 0.04	4.27 ± 0.06	4.30 ± 0.06
FastSpeech2	3.93 ± 0.05	3.30 ± 0.05	3.50 ± 0.05
DiffSinger	3.37 ± 0.05	2.90 ± 0.05	2.97 ± 0.05
DurIAN	2.97 ± 0.06	2.47 ± 0.05	2.20 ± 0.04
FT-GAN	$\mathbf{4.10 \pm 0.05}$	$\mathbf{3.70 \pm 0.06}$	$\mathbf{3.67 \pm 0.05}$

Table 4: MOS test result with 95% confidence interval of the ground truth and different models on Peking Opera synthesis.

Thanks